

C:\Users\stacey\Documents\MATLAB\advmath.m

```
1  Fs = 10000;    %samples per second
2  t = -0.25:1/Fs:0.5; %steps
3
4  %fourier series
5  N_terms = 50;  %number of harmonic terms that would be a triangle
6  Vm = zeros(size(t));
7
8
9  for n = 1:2:N_terms    %only the odd numbers are being looped
10     Vm = Vm + ((-1)^((n-1)/2) / n^2) * cos(n * 4 * pi * t);
11 end
12
13
14 Vm = Vm * (8/pi^2);
15
16 %Carrier
17 Vc = cos(20 * pi * t);
18
19 x = Vm .* Vc;    % x(t) = Vm(t).Vc(t)
20
21 N = length(x);    % total number of samples in x(t)
22 X = fft(x) / N;
23 f = (-N/2:N/2-1) * (Fs/N);
24 X_shift = fftshift(X);
25
26 figure;
27
28 %Time Domain
29 subplot(2,1,1);
30 plot(t, x, 'b', 'LineWidth', 1.5);
31 title('Time Domain Representation of Modulated Signal X(t)', 'FontWeight', 'bold');
32 xlabel('Time (s)');
33 ylabel('Amplitude');
34 ylim([-1.5 1.5]);
35 xlim([-0.25 0.5]);
36 grid on;
37
38 %Frequency Domain
39 subplot(2,1,2);
40 stem(f, abs(X_shift), 'b', 'filled', 'MarkerSize', 4, 'LineWidth', 0.8);
41 title('Frequency Domain Representation of X(t)', 'FontWeight', 'bold');
42 xlabel('Frequency (Hz)');
43 ylabel('Amplitude');
44 xlim([-60 70]);
45 ylim([0 0.55]);
46 grid on;
```

